THE COMPARISON MAP BETWEEN FIBRE PRODUCTS OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites one same-chapter lemma, delivered as a navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The diagonal of a morphism of formal algebraic spaces is representable, a monomorphism, locally quasi-finite, locally of finite type and separated. Same chapter, Lemma lemma-diagonal-morphism-formal-algebraic-spaces (statement lines 3548-3554), invoked at proof lines 6858-6859. Supplies every property asserted of i, by base change along X x_T Y -> Z x_T Z.
- Separated and quasi-separatedness of a morphism of formal algebraic spaces, read on the diagonal. Same chapter, Definition definition-separated and the separation discussion of the chapter. Turns 'Z -> T is separated' into the closedness of the diagonal and 'Z -> T is quasi-separated' into its quasi-compactness, which are the extra hypotheses of items (2) and (3).

Source reference. Original lemma 0AN2: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3549–3549. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Z$, $g : Y \to Z$ and $Z \to T$ be morphisms of formal algebraic spaces over S. Consider the induced morphism $i : X \times_Z Y \to X \times_T Y$. Then*

(1) *i is representable (by schemes), locally of finite type, locally quasi-finite, separated, and a monomorphism,*
(2) *if $Z \to T$ is separated, then i is a closed immersion, and*
(3) *if $Z \to T$ is quasi-separated, then i is quasi-compact.*

Proof. By general category theory the following diagram

$$\begin{array}{ccc} X \times_Z Y & \xrightarrow[i]{} & X \times_T Y \\ \downarrow & & \downarrow \\ Z & \xrightarrow{\Delta_{Z/T}} & Z \times_T Z \end{array}$$

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is a fibre product diagram. Hence i is the base change of the diagonal morphism $\Delta_{Z/T}$. Thus the lemma follows from Lemma 0AN2. $\square$